

Term 3 Practice Activity: C0 Discrete Probability Mass Functions

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Evaluated criteria

- C0.

1 General Problem

Problem. Let X be a discrete random variable that can take values $x_1, x_2, \dots, x_n$ with probability mass function

x	x_1	x_2	$\cdots$	x_n
$P(X = x)$	p_1	p_2	$\cdots$	p_n

where one or more probabilities may be unknown parameters.

Questions/Tasks. (a) Use the pmf conditions to determine the unknown parameter(s):

$$\sum_{i=1}^n p_i = 1 \quad \text{and} \quad p_i \geq 0 \text{ for all } i.$$

(b) Compute required probabilities such as $P(X = a)$, $P(X \leq a)$, $P(X \geq a)$, or $P(a \leq X \leq b)$ by adding the relevant pmf values. (c) Interpret each result as a percentage in context.

C0

(a) Set up the total-probability equation

$$p_1 + p_2 + \cdots + p_n = 1$$

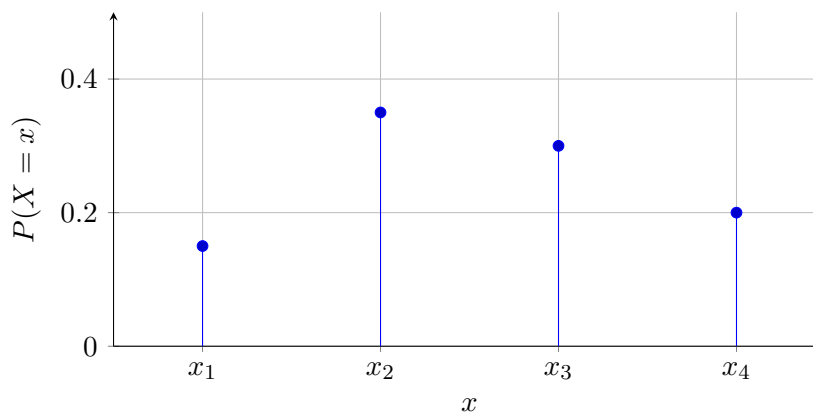
and solve for the unknown parameter(s). Then check nonnegativity of every probability.

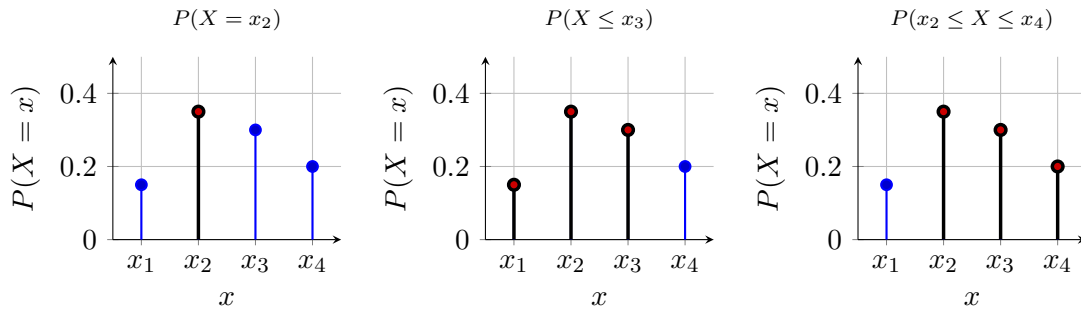
(b) For any event A on the support,

$$P(X \in A) = \sum_{x_i \in A} P(X = x_i).$$

In a pmf stem graph, each horizontal value x_i is a possible value of X , and the vertical stem height is $P(X = x_i)$. The full graph shows all stems $\{(x_i, p_i)\}_{i=1}^n$, while highlighted stems show the values included in an event such as $P(X = a)$, $P(X \leq b)$, or $P(a \leq X \leq b)$.

$$P(X = x) = \{(x_1, p_1), (x_2, p_2), \dots, (x_n, p_n)\}.$$





So exact, at most, at least, and interval probabilities are found by summing the corresponding highlighted pmf entries.

(c) Convert each probability to a percentage by multiplying by 100%.

2 App Notifications per Hour

Problem. Let X be the number of app notifications a student receives in one hour. The probability mass function is

x	0	1	2	3
$P(X = x)$	0.2	0.3	k	0.1

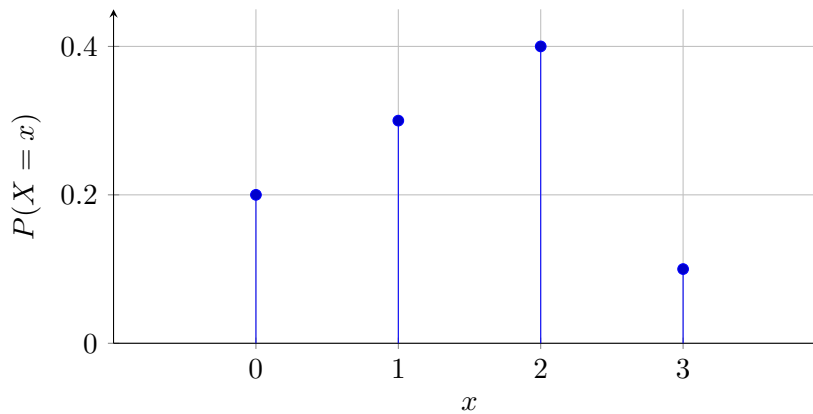
Questions/Tasks. (a) Determine k so that this is a valid probability distribution. (b) Compute: (i) $P(X = 2)$, (ii) $P(X \leq 2)$, and (iii) $P(1 \leq X \leq 2)$. (c) Interpret these results in terms of percentage probability.

C0

(a) For a valid pmf,

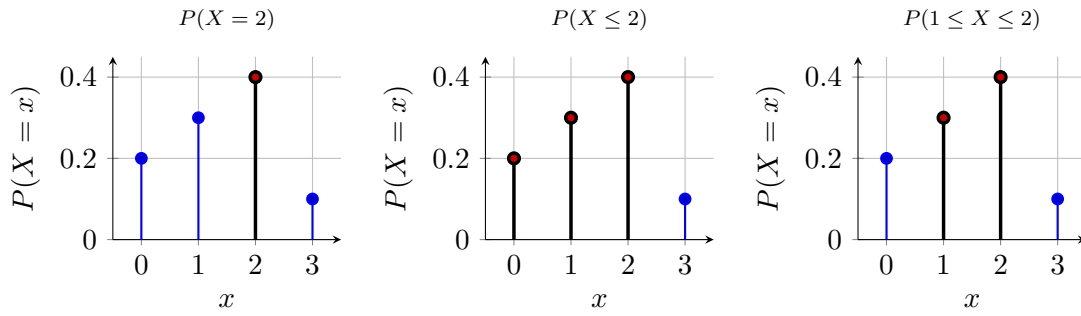
$$0.2 + 0.3 + k + 0.1 = 1 \Rightarrow k = 0.4.$$

$$P(X = x) = \{(0, 0.2), (1, 0.3), (2, 0.4), (3, 0.1)\}.$$



(b)

$$P(X = 2) = 0.4, \quad P(X \leq 2) = 0.2 + 0.3 + 0.4 = 0.9, \quad P(1 \leq X \leq 2) = 0.3 + 0.4 = 0.7.$$



(c) There is a 40% probability of exactly 2 notifications, a 90% probability of at most 2 notifications, and a 70% probability of between 1 and 2 notifications.

3 Defective Items in a Batch

Problem. A quality check records Y , the number of defective items in a small batch. Suppose

y	0	1	2	3
$P(Y = y)$	0.5	c	0.2	0.1

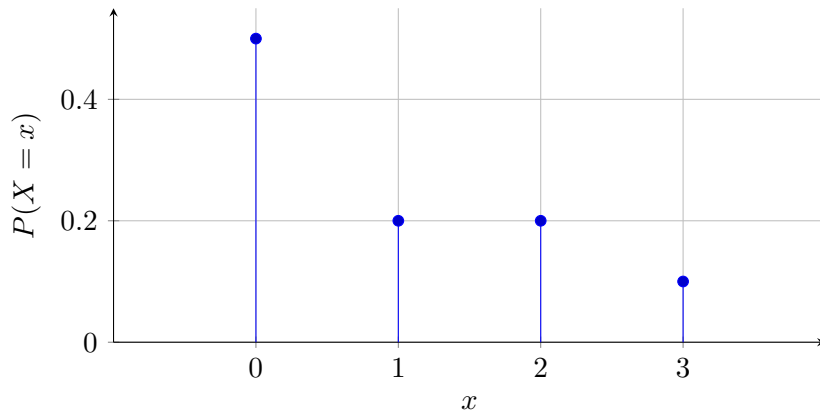
Questions/Tasks. (a) Determine c so that the distribution is valid. (b) Compute: (i) $P(Y = 1)$, (ii) $P(1 \leq Y \leq 2)$, and (iii) $P(Y \geq 1)$. (c) Interpret these results in terms of percentage probability.

C0

(a) Sum-to-1 gives

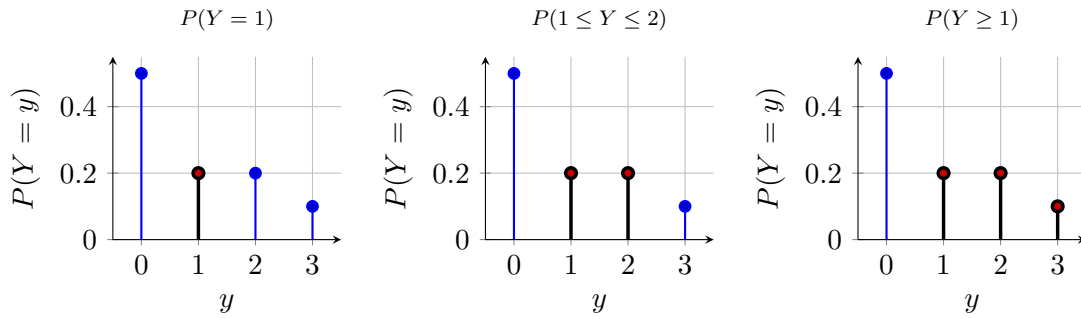
$$0.5 + c + 0.2 + 0.1 = 1 \Rightarrow c = 0.2.$$

$$P(Y = y) = \{(0, 0.5), (1, 0.2), (2, 0.2), (3, 0.1)\}.$$



(b)

$$P(Y = 1) = 0.2, \quad P(1 \leq Y \leq 2) = 0.2 + 0.2 = 0.4, \quad P(Y \geq 1) = 0.2 + 0.2 + 0.1 = 0.5.$$



(c) There is a 20% probability of exactly 1 defective item, a 40% probability of 1 or 2 defective items, and a 50% probability of at least 1 defective item.

4 Lost Pens in a Small Box

Problem. Let Z be the number of lost pens found in a small box. The probability mass function is

z	0	1	2	3
$P(Z = z)$	0.4	0.5	a	0.2

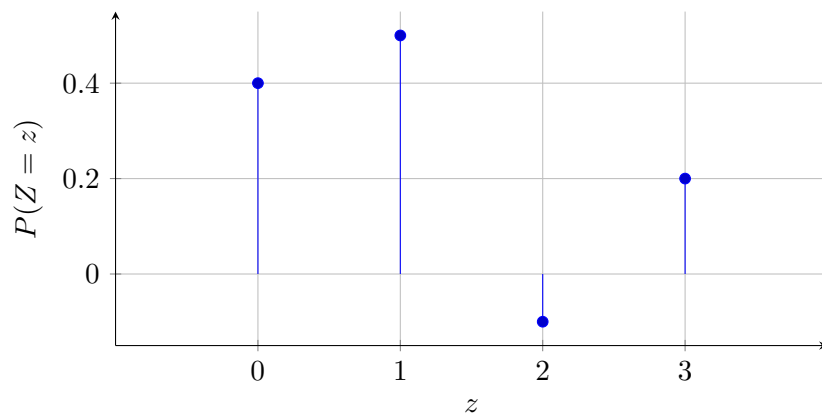
Questions/Tasks. (a) Solve for a so the probabilities sum to 1. (b) Decide whether the resulting distribution is valid. (c) Using the computed a , compute $P(Z \geq 2)$.

C0

(a) From total probability,

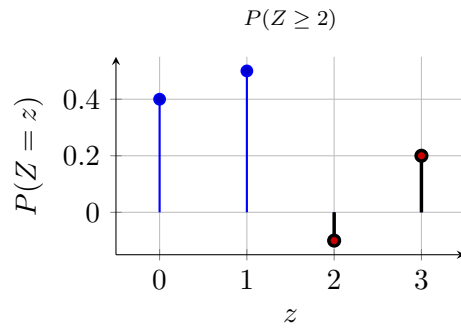
$$0.4 + 0.5 + a + 0.2 = 1 \Rightarrow 1.1 + a = 1 \Rightarrow a = -0.1.$$

$$P(Z = z) = \{(0, 0.4), (1, 0.5), (2, -0.1), (3, 0.2)\}.$$



(b)

$$P(Z \geq 2) = P(Z = 2) + P(Z = 3) = a + 0.2 = -0.1 + 0.2 = 0.1.$$



(c) Not a valid probability distribution because $a < 0$.

5 Late Deliveries in One Day

Problem. Let W be the number of late deliveries made by a shop in one day. The probability mass function is

w	0	1	2	3
$P(W = w)$	0.1	b	$2b$	0.3

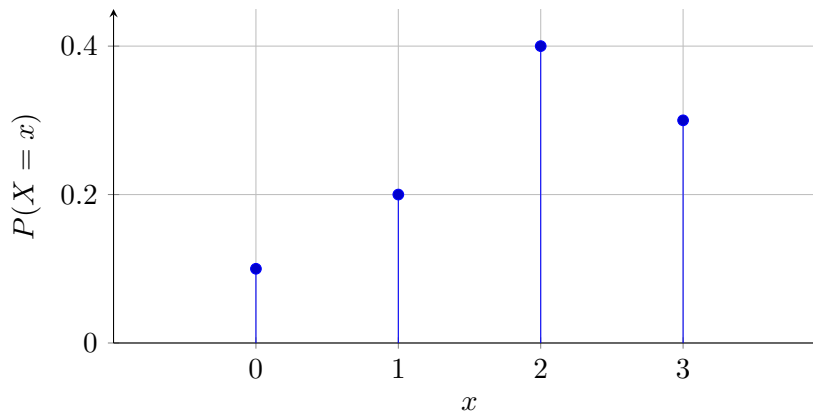
Questions/Tasks. (a) Determine b so that the distribution is valid. (b) Compute: (i) $P(W = 3)$, (ii) $P(W \geq 2)$, and (iii) $P(1 \leq W \leq 3)$. (c) Interpret these results in terms of percentage probability.

C0

(a) From total probability,

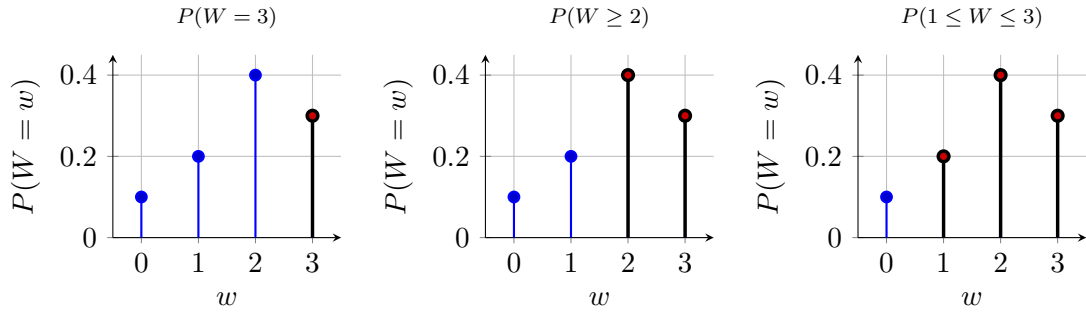
$$0.1 + b + 2b + 0.3 = 1 \Rightarrow 3b = 0.6 \Rightarrow b = 0.2.$$

$$P(W = w) = \{(0, 0.1), (1, 0.2), (2, 0.4), (3, 0.3)\}.$$



(b)

$$P(W = 3) = 0.3, \quad P(W \geq 2) = 0.4 + 0.3 = 0.7, \quad P(1 \leq W \leq 3) = 0.2 + 0.4 + 0.3 = 0.9.$$



(c) There is a 30% probability of exactly 3 late deliveries, a 70% probability of at least 2 late deliveries, and a 90% probability of between 1 and 3 late deliveries.

6 Customers in a 10-Minute Window

Problem. A café tracks T , the number of customers arriving in a 10-minute window.

t	1	2	3	4
$P(T = t)$	λ	$0.7 - \lambda$	0.2	0.1

Questions/Tasks. (a) Determine λ so that this is a valid probability distribution. (b) Compute: (i) $P(T = 2)$, (ii) $P(T \neq 4)$, and (iii) $P(2 \leq T \leq 3)$. (c) Interpret these results in terms of percentage probability.

C0

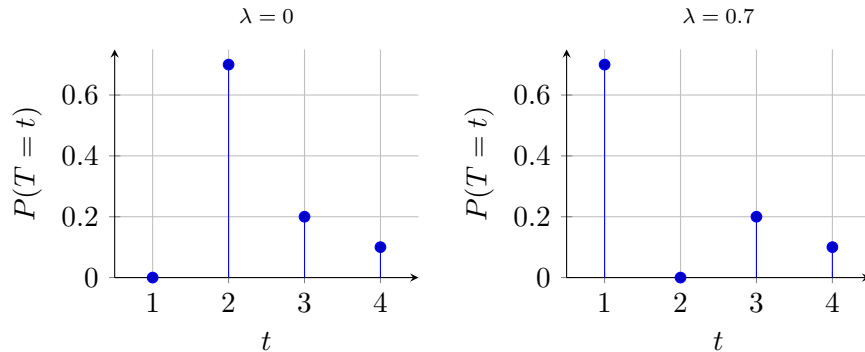
(a) Sum-to-1 gives

$$\lambda + (0.7 - \lambda) + 0.2 + 0.1 = 1,$$

which is true for all λ . For a valid pmf we need nonnegative probabilities:

$$\lambda \geq 0, \quad 0.7 - \lambda \geq 0 \Rightarrow 0 \leq \lambda \leq 0.7.$$

$$P(T = t) = \{(1, \lambda), (2, 0.7 - \lambda), (3, 0.2), (4, 0.1)\}, \quad 0 \leq \lambda \leq 0.7.$$



(b)

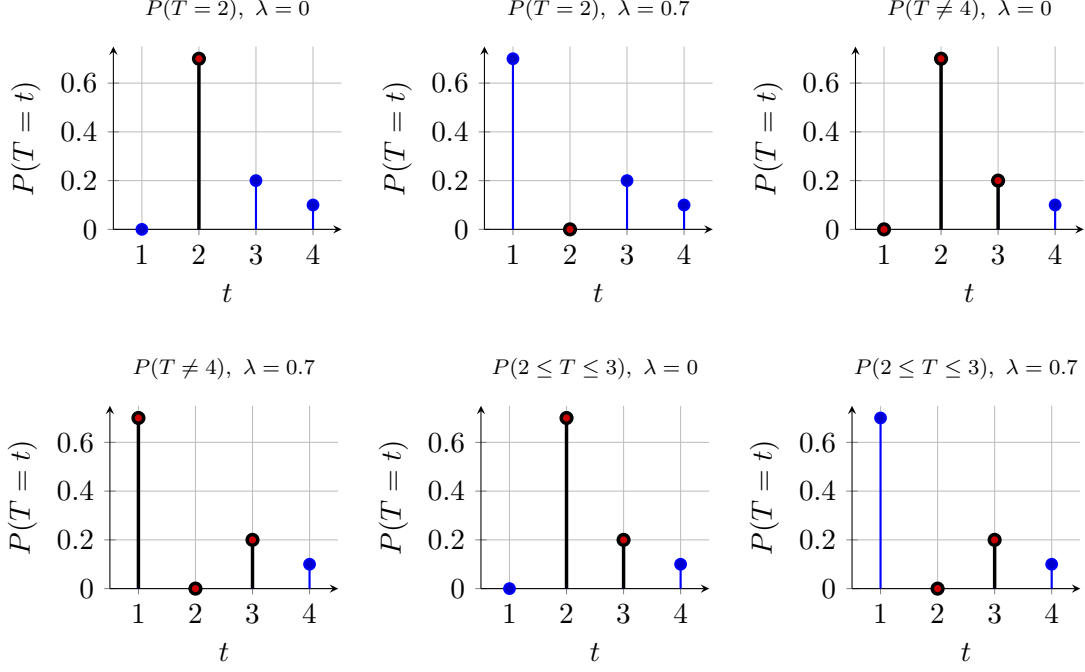
$$P(T = 2) = 0.7 - \lambda, \quad P(T \neq 4) = 1 - 0.1 = 0.9, \quad P(2 \leq T \leq 3) = (0.7 - \lambda) + 0.2 = 0.9 - \lambda.$$

At the lower extreme $\lambda = 0$:

$$P(T = 2) = 0.7, \quad P(T \neq 4) = 0.9, \quad P(2 \leq T \leq 3) = 0.9.$$

At the upper extreme $\lambda = 0.7$:

$$P(T = 2) = 0, \quad P(T \neq 4) = 0.9, \quad P(2 \leq T \leq 3) = 0.2.$$



(c) There is a $100(0.7 - \lambda)\%$ probability of exactly 2 customers, a 90% probability of a number of customers different from 4, and a $100(0.9 - \lambda)\%$ probability of between 2 and 3 customers.

7 Customers in a Short Checkout Window

Problem. A café tracks X , the number of customers arriving in a short checkout window.

x	1	2	3	4
$P(X = x)$	0.6	d	0.5	d

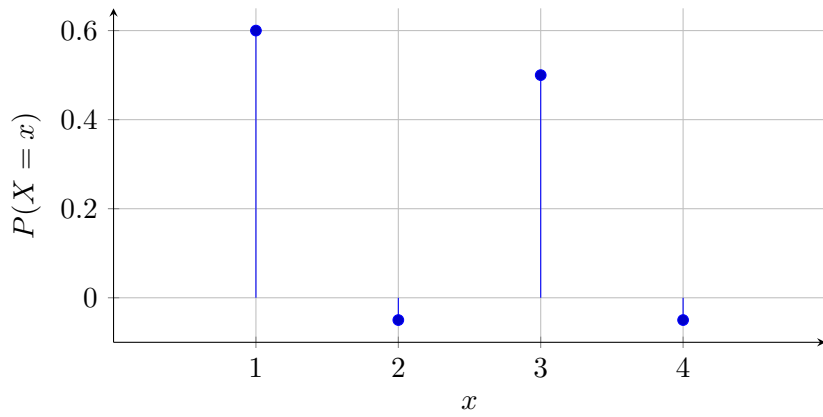
Questions/Tasks. (a) Solve for d so the probabilities sum to 1. (b) Decide whether the resulting distribution is valid. (c) Using the computed d , compute $P(2 \leq X \leq 4)$.

C0

(a) Using total probability,

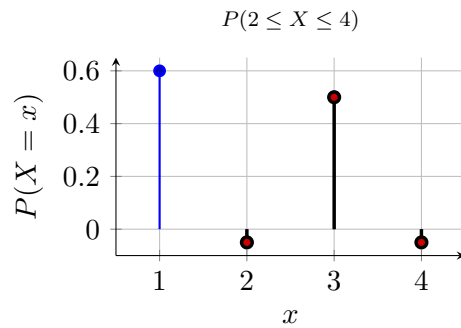
$$0.6 + d + 0.5 + d = 1 \Rightarrow 1.1 + 2d = 1 \Rightarrow 2d = -0.1 \Rightarrow d = -0.05.$$

$$P(X = x) = \{(1, 0.6), (2, -0.05), (3, 0.5), (4, -0.05)\}.$$



(b)

$$P(2 \leq X \leq 4) = P(X=2) + P(X=3) + P(X=4) = d + 0.5 + d = -0.05 + 0.5 - 0.05 = 0.4.$$



(c) Not a valid probability distribution because $d < 0$.

8 Correct Answers on a Short Quiz

Problem. Let Y be the number of correct answers on a 4-question quiz. The pmf is

$$P(Y=y) = \begin{cases} r, & y=0, \\ 2r, & y=1, \\ 3r, & y=2, \\ 2r, & y=3, \\ 1-8r, & y=4. \end{cases}$$

Questions/Tasks. (a) Determine all values of r that make this a valid distribution. (b) Compute: (i) $P(Y=4)$, (ii) $P(Y \geq 3)$, and (iii) $P(1 \leq Y \leq 3)$. (c) Interpret these results in terms of percentage probability.

C0

(a) From the total probability condition,

$$r + 2r + 3r + 2r + (1 - 8r) = 1,$$

which is true for all r .

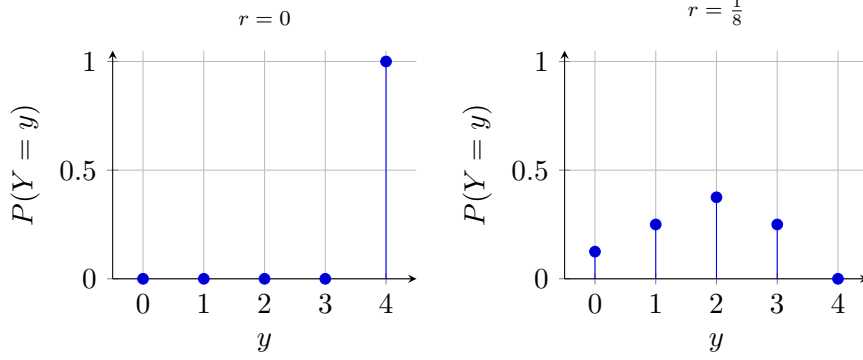
For the distribution to be valid, all probabilities must be nonnegative:

$$r \geq 0, \quad 1 - 8r \geq 0.$$

Hence,

$$0 \leq r \leq \frac{1}{8}.$$

$$P(Y = y) = \{(0, r), (1, 2r), (2, 3r), (3, 2r), (4, 1 - 8r)\}, \quad 0 \leq r \leq \frac{1}{8}.$$



(b)

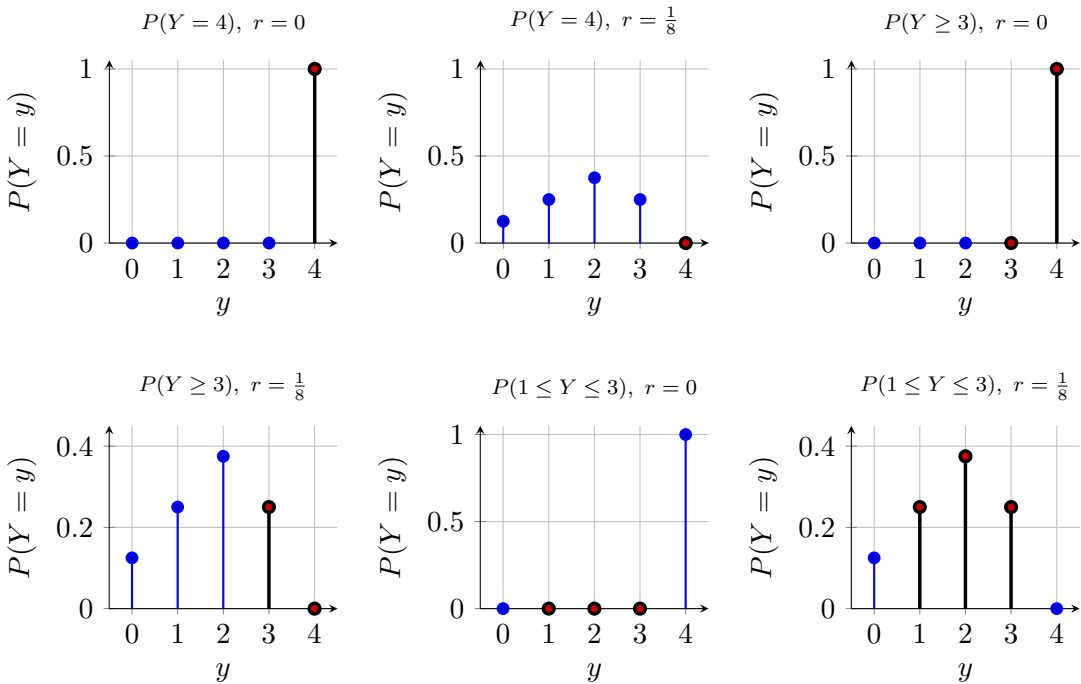
$$P(Y = 4) = 1 - 8r, \quad P(Y \geq 3) = 2r + (1 - 8r) = 1 - 6r, \quad P(1 \leq Y \leq 3) = 2r + 3r + 2r = 7r.$$

At the lower extreme $r = 0$:

$$P(Y = 4) = 1, \quad P(Y \geq 3) = 1, \quad P(1 \leq Y \leq 3) = 0.$$

At the upper extreme $r = \frac{1}{8}$:

$$P(Y = 4) = 0, \quad P(Y \geq 3) = \frac{1}{4}, \quad P(1 \leq Y \leq 3) = \frac{7}{8}.$$



(c) The percentage probabilities are $100(1 - 8r)\%$ for exactly 4 correct answers, $100(1 - 6r)\%$ for at least 3 correct answers, and $700r\%$ for between 1 and 3 correct answers.

9 Messages Received in an Hour

Problem. Let Z be the number of messages received in one hour. Suppose

z	0	1	2	3	4
$P(Z = z)$	m	$2m$	$3m$	$2m$	m

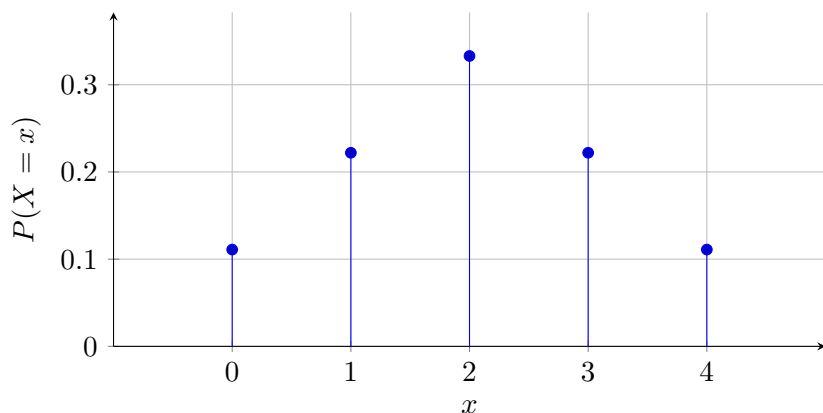
Questions/Tasks. (a) Find m so this is a valid probability distribution. (b) Compute: (i) $P(Z = 2)$, (ii) $P(Z \geq 3)$, and (iii) $P(1 \leq Z \leq 3)$. (c) Interpret these results in terms of percentage probability.

C0

(a) Use the sum-to-1 rule:

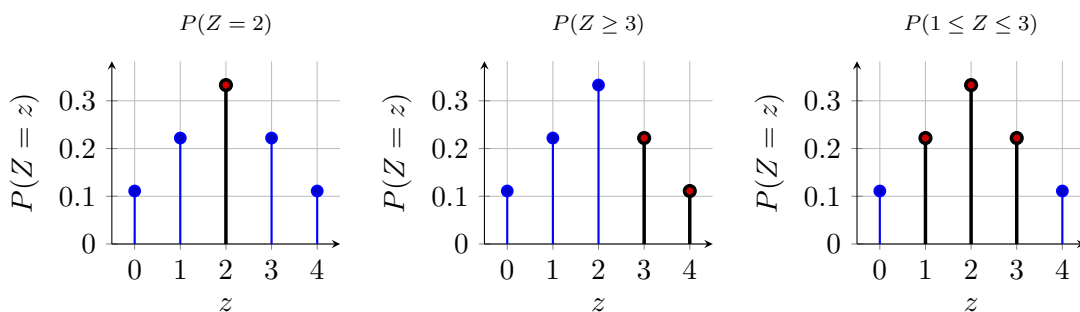
$$m + 2m + 3m + 2m + m = 9m = 1 \Rightarrow m = \frac{1}{9}.$$

$$P(Z = z) = \{(0, 0.111), (1, 0.222), (2, 0.333), (3, 0.222), (4, 0.111)\}.$$



(b)

$$P(Z = 2) = 3m = \frac{1}{3}, \quad P(Z \geq 3) = 2m + m = 3m = \frac{1}{3}, \quad P(1 \leq Z \leq 3) = 2m + 3m + 2m = 7m = \frac{7}{9}.$$



(c) There is a 33.3% probability of exactly 2 messages, a 33.3% probability of at least 3 messages, and a 77.7% probability of between 1 and 3 messages.

10 Calls Received in a Busy Hour

Problem. Let W be the number of calls received in a busy hour, with

w	0	1	2	3	4
$P(W = w)$	h	0.4	$2h$	0.3	$0.3 - 3h$

Questions/Tasks. (a) Verify that the probabilities sum to 1. (b) Find the values of h for which this is a valid distribution. (c) For valid h , compute $P(W \geq 2)$.

C0

(a)

$$h + 0.4 + 2h + 0.3 + (0.3 - 3h) = 1,$$

so the probabilities sum to 1 for all h .

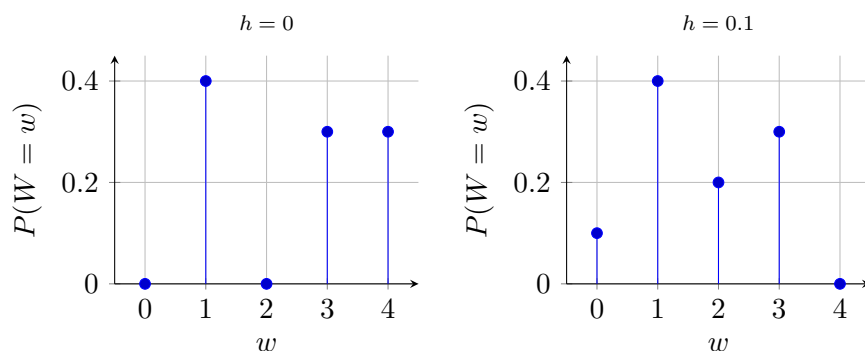
(b) Validity requires nonnegative probabilities:

$$h \geq 0, \quad 2h \geq 0, \quad 0.3 - 3h \geq 0.$$

From $0.3 - 3h \geq 0$, we get $0.3 \geq 3h$, so $0.1 \geq h$, that is, $h \leq 0.1$. Hence

$$0 \leq h \leq 0.1.$$

$$P(W = w) = \{(0, h), (1, 0.4), (2, 2h), (3, 0.3), (4, 0.3 - 3h)\}, \quad 0 \leq h \leq 0.1.$$



(c)

$$P(W \geq 2) = P(W = 2) + P(W = 3) + P(W = 4) = 2h + 0.3 + (0.3 - 3h) = 0.6 - h.$$

Thus,

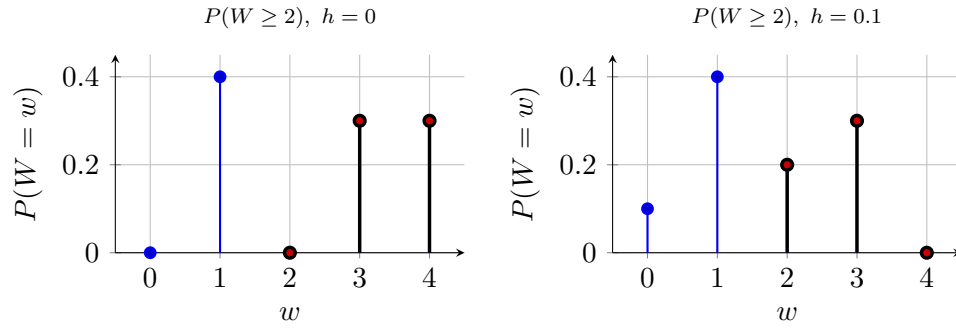
$$P(W \geq 2) = 0.6 - h, \quad 0 \leq h \leq 0.1.$$

At the lower extreme $h = 0$:

$$P(W \geq 2) = 0.6.$$

At the upper extreme $h = 0.1$:

$$P(W \geq 2) = 0.5.$$



11 Goals in a Match Summary

Problem. Let T be the number of goals scored by a team in a match summary. The probability mass function is

$$P(T = t) = \begin{cases} p, & t = 0, \\ \frac{3}{4} - 2p, & t = 1, \\ p, & t = 2, \\ \frac{1}{4}, & t = 3. \end{cases}$$

Questions/Tasks. (a) Determine all valid values of p . Since all listed probabilities must satisfy nonnegativity, require

$$p \geq 0, \quad \frac{3}{4} - 2p \geq 0,$$

so the feasible interval is

$$0 \leq p \leq \frac{3}{8}.$$

(b) Compute: (i) $P(T = 1)$, (ii) $P(T \in \{0, 2, 3\})$, and (iii) $P(1 \leq T \leq 3)$. (c) Interpret these results in terms of percentage probability.

C0

(a) The sum condition is

$$p + \left(\frac{3}{4} - 2p\right) + p + \frac{1}{4} = 1,$$

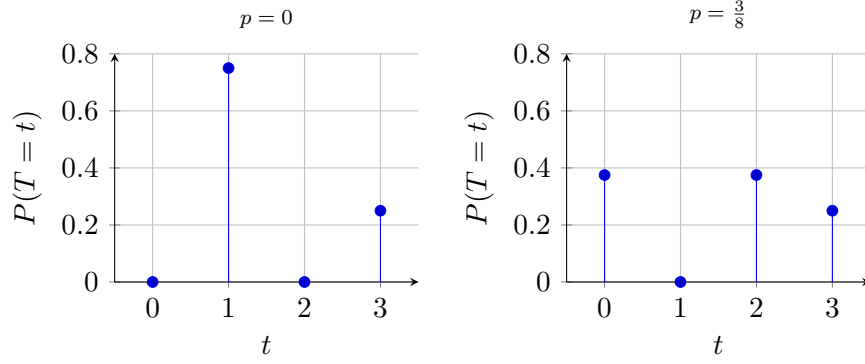
which is true for all p . For the pmf to be valid, each probability must be nonnegative:

$$p \geq 0, \quad \frac{3}{4} - 2p \geq 0.$$

Hence

$$0 \leq p \leq \frac{3}{8}.$$

$$P(T = t) = \{(0, p), (1, \frac{3}{4} - 2p), (2, p), (3, \frac{1}{4})\}, \quad 0 \leq p \leq \frac{3}{8}.$$



(b)

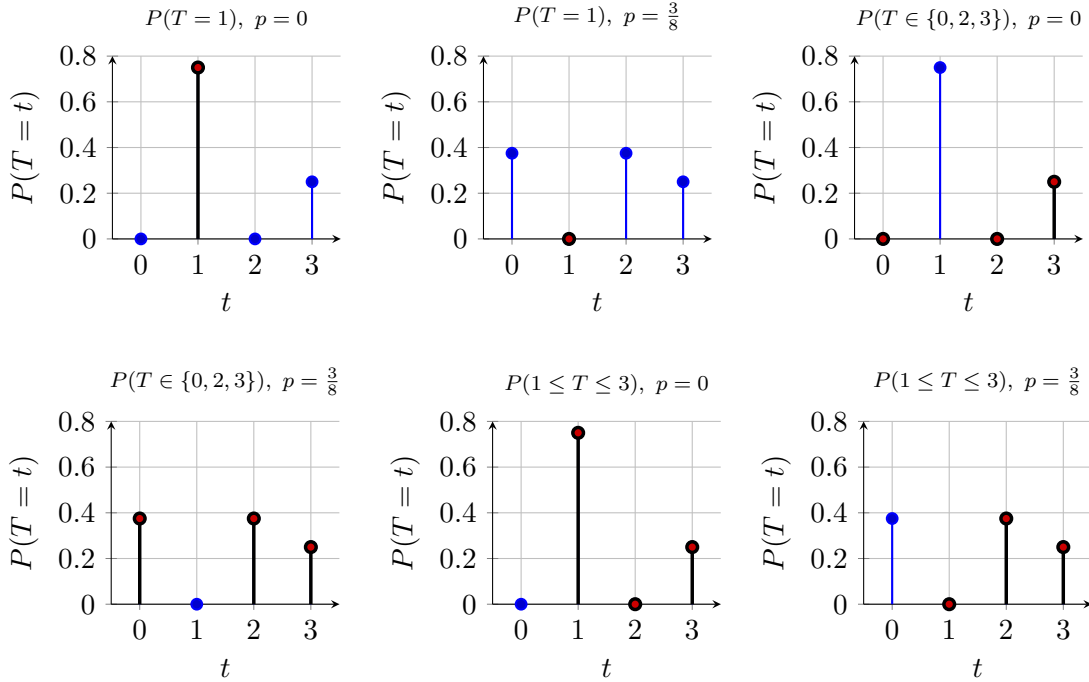
$$P(T=1) = \frac{3}{4} - 2p, \quad P(T \in \{0, 2, 3\}) = p + p + \frac{1}{4} = 2p + \frac{1}{4}, \quad P(1 \leq T \leq 3) = 1 - p.$$

At the lower extreme $p = 0$:

$$P(T=1) = \frac{3}{4}, \quad P(T \in \{0, 2, 3\}) = \frac{1}{4}, \quad P(1 \leq T \leq 3) = 1.$$

At the upper extreme $p = \frac{3}{8}$:

$$P(T=1) = 0, \quad P(T \in \{0, 2, 3\}) = 1, \quad P(1 \leq T \leq 3) = \frac{5}{8}.$$



(c) The percentage probabilities are $100 \left(\frac{3}{4} - 2p \right) \%$ for exactly 1 goal, $100 \left(2p + \frac{1}{4} \right) \%$ for 0, 2, or 3 goals, and $100(1-p)\%$ for between 1 and 3 goals.

12 Late Buses in a Morning

Problem. Let X be the number of late buses observed in a morning, with

x	0	1	2	3
$P(X=x)$	u	v	$2u$	v

Questions/Tasks. (a) Find the values of u and v for which this is a valid distribution. (b) Compute (i) $P(X = 2)$, (ii) $P(X > 1)$, and (iii) $P(1 \leq X \leq 3)$. (c) Interpret the results as percentages.

C0

(a)

$$u + v + 2u + v = 1 \Rightarrow 3u + 2v = 1.$$

So

$$v = \frac{1 - 3u}{2}.$$

Validity also requires

$$u \geq 0, \quad v \geq 0.$$

From

$$\frac{1 - 3u}{2} \geq 0,$$

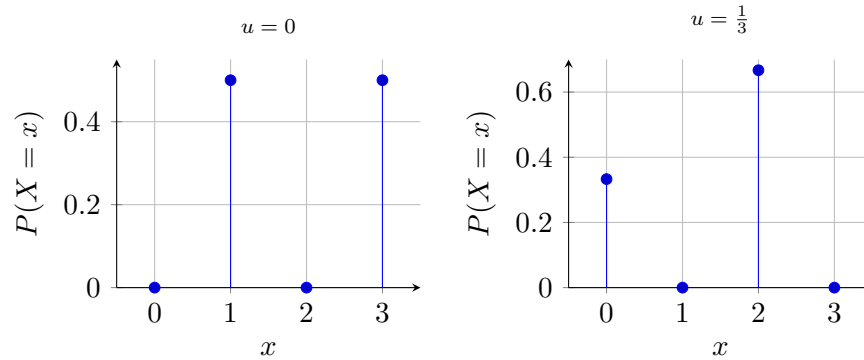
we get

$$1 - 3u \geq 0 \Rightarrow u \leq \frac{1}{3}.$$

Hence

$$0 \leq u \leq \frac{1}{3}, \quad v = \frac{1 - 3u}{2}.$$

$$P(X = x) = \{(0, u), (1, v), (2, 2u), (3, v)\}, \quad 0 \leq u \leq \frac{1}{3}, \quad v = \frac{1 - 3u}{2}.$$



(b)

$$P(X = 2) = 2u.$$

Also,

$$P(X > 1) = P(X = 2) + P(X = 3) = 2u + v = 2u + \frac{1 - 3u}{2} = \frac{1 + u}{2}.$$

And

$$P(1 \leq X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3) = v + 2u + v = 2v + 2u = 1 - u.$$

At the lower extreme $u = 0$ (so $v = \frac{1}{2}$):

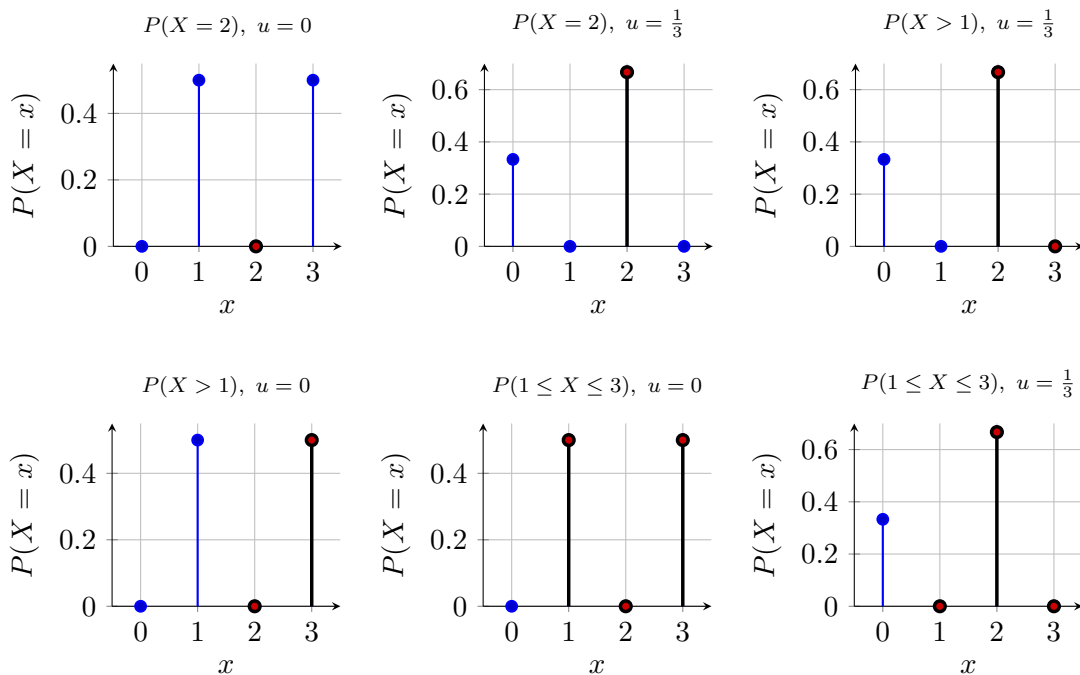
$$P(X = 2) = 0, \quad P(X > 1) = \frac{1}{2}, \quad P(1 \leq X \leq 3) = 1.$$

At the upper extreme $u = \frac{1}{3}$ (so $v = 0$):

$$P(X = 2) = \frac{2}{3}, \quad P(X > 1) = \frac{2}{3}, \quad P(1 \leq X \leq 3) = \frac{2}{3}.$$

(c)

$$P(X = 2) = 2u, \quad P(X > 1) = \frac{1+u}{2}, \quad P(1 \leq X \leq 3) = 1 - u.$$



So there is a $100(2u)\%$ probability of exactly 2 late buses, a $100\left(\frac{1+u}{2}\right)\%$ probability of more than 1 late bus, and a $100(1-u)\%$ probability of between 1 and 3 late buses.

13 Deliveries Delayed in a Morning

Problem. Let Y be the number of deliveries delayed in a morning. The probability mass function is

y	0	1	2	3
$P(Y=y)$	0.3	γ	γ	γ

and it is known that

$$P(Y = 2) = 2P(Y = 0).$$

Questions/Tasks. (a) Solve for γ so the probabilities sum to 1. (b) Decide whether the resulting distribution is valid. (c) Using the computed γ , compute $P(Y \geq 1)$.

C0

(a) From the added condition,

$$\gamma = 2P(Y = 0) = 2(0.3) = 0.6.$$

Using total probability,

$$0.3 + \gamma + \gamma + \gamma = 1.$$

Substituting $\gamma = 0.6$ gives

$$0.3 + 0.6 + 0.6 + 0.6 = 2.1 \neq 1.$$

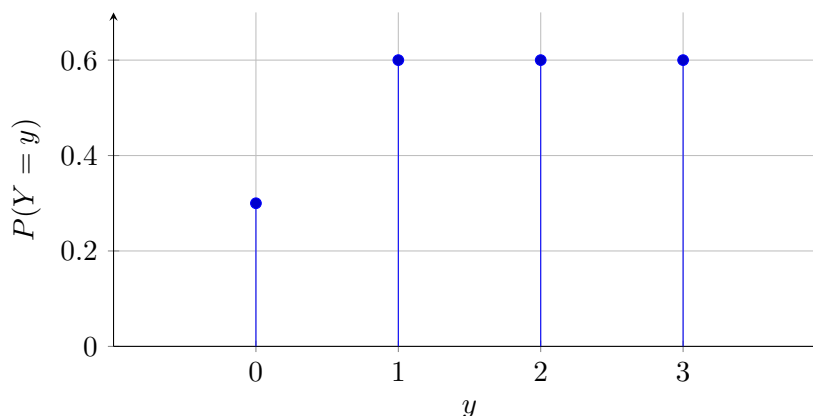
Therefore, the conditions are incompatible with the requirement that probabilities sum to 1, so this does not produce a valid probability mass function.

Equivalently, if we force the sum-to-1 condition first, then

$$0.3 + 3\gamma = 1 \Rightarrow 3\gamma = 0.7 \Rightarrow \gamma = \frac{7}{30},$$

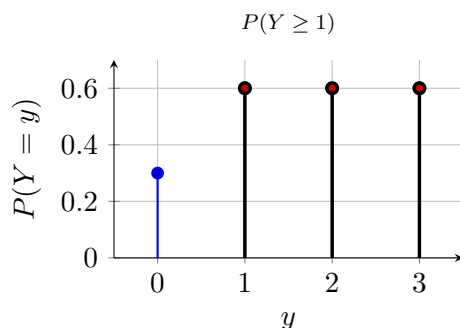
but this contradicts the required condition $\gamma = 0.6$.

$$P(Y = y) = \{(0, 0.3), (1, 0.6), (2, 0.6), (3, 0.6)\}.$$



(b)

$$P(Y \geq 1) = P(Y = 1) + P(Y = 2) + P(Y = 3) = \gamma + \gamma + \gamma = 3\gamma = 3(0.6) = 1.8.$$



(c) **Not a valid probability distribution** because the probabilities cannot simultaneously satisfy both the sum-to-1 condition and the condition $P(Y = 2) = 2P(Y = 0)$.

If the value from the added condition is used, then

14 Customers in a Short Window

Problem. A small store records Z , the number of customers arriving in a short window. Suppose

z	0	1	2	3	4
$P(Z = z)$	s	$2s$	$4s$	$2s$	s

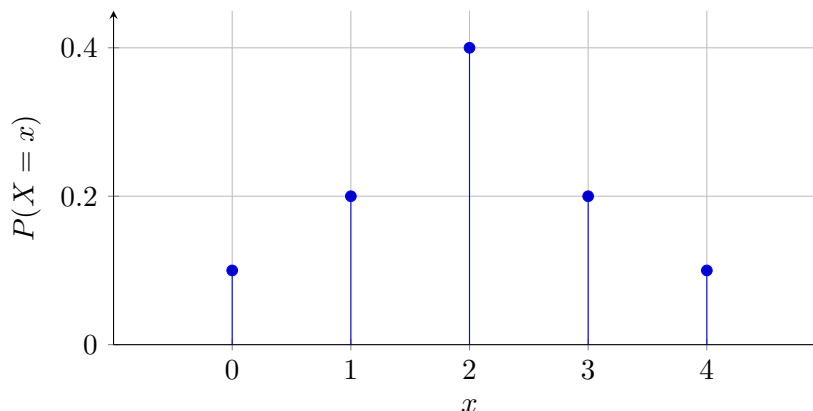
Questions/Tasks. (a) Determine s . (b) Compute: (i) $P(Z = 2)$, (ii) $P(1 \leq Z \leq 3)$, and (iii) $P(Z \geq 2)$. (c) Interpret these results in terms of percentage probability.

C0

(a) Use total probability 1:

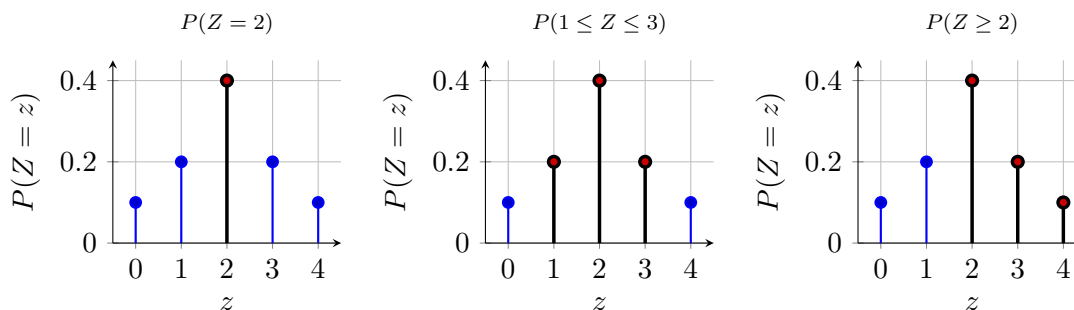
$$s + 2s + 4s + 2s + s = 10s = 1 \Rightarrow s = \frac{1}{10}.$$

$$P(Z = z) = \{(0, 0.1), (1, 0.2), (2, 0.4), (3, 0.2), (4, 0.1)\}.$$



(b)

$$P(Z = 2) = 4s = \frac{2}{5}, \quad P(1 \leq Z \leq 3) = 2s + 4s + 2s = 8s = \frac{4}{5}, \quad P(Z \geq 2) = 4s + 2s + s = 7s = \frac{7}{10}.$$



(c) There is a 40% probability of exactly 2 customers, an 80% probability of between 1 and 3 customers, and a 70% probability of at least 2 customers.

15 Correct Answers on a 3-Question Check

Problem. Let W be the number of correct answers on a 3-question check.

w	0	1	2	3
$P(W = w)$	$2q$	q	$\frac{1}{2} - q$	$\frac{1}{4}$

Questions/Tasks. (a) Determine the range of values of q so that this is a valid distribution, and verify the value of q obtained from the sum-to-1 condition. (b) Compute: (i) $P(W = 3)$, (ii) $P(W \in \{0, 2\})$, and (iii) $P(1 \leq W \leq 2)$. (c) Interpret these results in terms of percentage probability.

C0

(a) For the distribution to be valid, all probabilities must be nonnegative:

$$2q \geq 0, \quad q \geq 0, \quad \frac{1}{2} - q \geq 0, \quad \frac{1}{4} \geq 0.$$

Hence the valid range is

$$0 \leq q \leq \frac{1}{2}.$$

Sum-to-1 gives

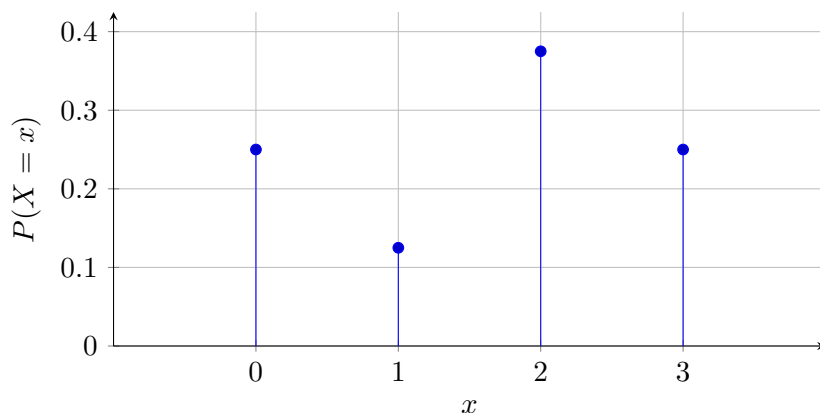
$$2q + q + \left(\frac{1}{2} - q\right) + \frac{1}{4} = 1 \Rightarrow 2q + \frac{3}{4} = 1 \Rightarrow q = \frac{1}{8}.$$

Since

$$0 \leq \frac{1}{8} \leq \frac{1}{2},$$

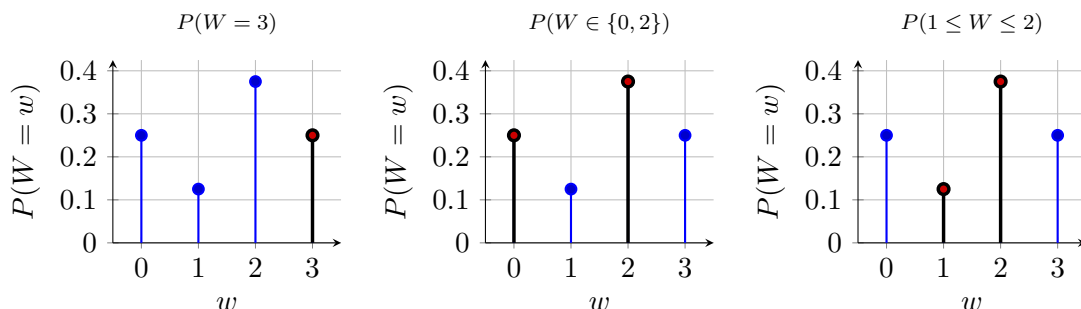
the value $q = \frac{1}{8}$ is valid.

$$P(W = w) = \{(0, 0.25), (1, 0.125), (2, 0.375), (3, 0.25)\}.$$



(b)

$$P(W = 3) = \frac{1}{4}, \quad P(W \in \{0, 2\}) = 2q + \left(\frac{1}{2} - q\right) = \frac{5}{8}, \quad P(1 \leq W \leq 2) = q + \left(\frac{1}{2} - q\right) = \frac{1}{2}.$$



(c) There is a 25% probability of exactly 3 correct answers, a 62.5% probability of getting 0 or 2 correct answers, and a 50% probability of getting 1 or 2 correct answers.

16 Scores in a Quick Check

Problem. Let T be the number of correct answers on a quick 3-question check.

t	0	1	2	3
$P(T = t)$	e	f	0.6	0.2

Questions/Tasks. (a) Solve for e and f so the probabilities satisfy the conditions

$$e = f - 0.1$$

(b) Decide whether the resulting distribution is valid. (c) Using the computed values, compute $P(T \leq 2)$.

C0

(a) From total probability,

$$e + f + 0.6 + 0.2 = 1 \Rightarrow e + f = 0.2.$$

Using the extra condition

$$e = f - 0.1.$$

Substitute into the first equation:

$$(f - 0.1) + f = 0.2 \Rightarrow 2f = 0.3 \Rightarrow f = 0.15.$$

Then

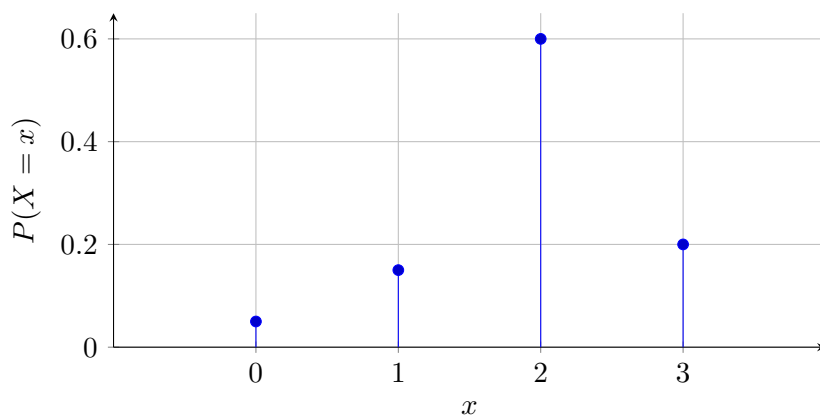
$$e = f - 0.1 = 0.05.$$

This is a valid probability distribution because

$$e = 0.05 \geq 0, \quad f = 0.15 \geq 0,$$

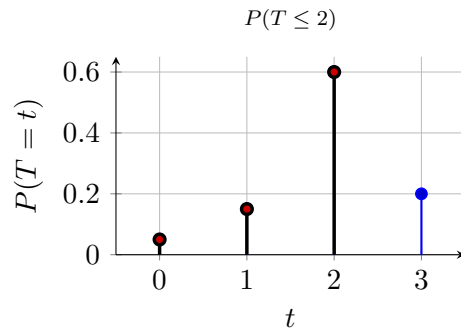
and the probabilities sum to 1.

$$P(T = t) = \{(0, 0.05), (1, 0.15), (2, 0.6), (3, 0.2)\}.$$



(b)

$$P(T \leq 2) = P(T = 0) + P(T = 1) + P(T = 2) = e + f + 0.6 = 0.05 + 0.15 + 0.6 = 0.8.$$



(c) There is an 80% probability of getting at most 2 correct answers.

17 Yellow Cards in a Match

Problem. Let X be the number of yellow cards shown in a match summary. The probability mass function is

x	0	1	2	3	4
$P(X=x)$	b	a	$a+b$	$2a$	a

and it is known that $P(X=4) = 4P(X=0)$.

Questions/Tasks. (a) Find a and b so the model is valid. (b) Compute: (i) $P(X=4)$, (ii) $P(X \leq 1)$, and (iii) $P(2 \leq X \leq 4)$. (c) Interpret these results in terms of percentage probability.

C0

(a) From the condition $P(X=4) = 4P(X=0)$,

$$a = 4b.$$

From total probability,

$$b + a + (a + b) + 2a + a = 1 \Rightarrow 2b + 5a = 1.$$

Substitute $a = 4b$:

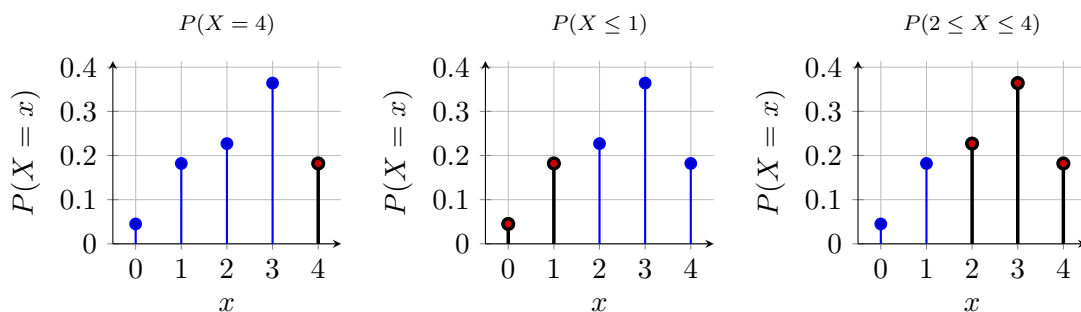
$$2b + 5(4b) = 1 \Rightarrow 22b = 1 \Rightarrow b = \frac{1}{22}.$$

Then

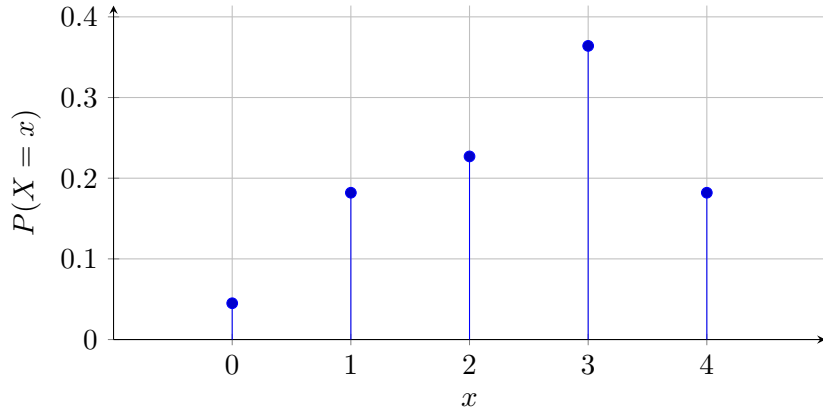
$$a = 4b = \frac{4}{22} = \frac{2}{11}.$$

(b)

$$P(X=4) = a = \frac{2}{11}, \quad P(X \leq 1) = b + a = \frac{1}{22} + \frac{2}{11} = \frac{5}{22}, \quad P(2 \leq X \leq 4) = (a+b) + 2a + a = b + 4a = \frac{1}{22} + \frac{8}{11} = \frac{17}{22}.$$



$$P(X = x) = \{(0, 0.045), (1, 0.182), (2, 0.227), (3, 0.364), (4, 0.182)\}.$$



(c) There is a $\frac{2}{11}$ probability of exactly 4 yellow cards, a $\frac{5}{22}$ probability of at most 1 yellow card, and a $\frac{17}{22}$ probability of between 2 and 4 yellow cards.

18 Defects Found in Final Check

Problem. Let Y be the number of defects found in a final check. Suppose

$$P(Y = y) = \begin{cases} r, & y = 0, \\ 2r, & y = 1, \\ \frac{1}{2} - 2r, & y = 2, \\ r, & y = 3, \\ \frac{1}{4}, & y = 4. \end{cases}$$

Questions/Tasks. (a) Determine the range of valid values of r and verify the value of r obtained from the sum-to-1 condition. (b) Compute: (i) $P(Y = 4)$, (ii) $P(Y \in \{0, 2, 4\})$, and (iii) $P(1 \leq Y \leq 3)$. (c) Interpret these results in terms of percentage probability.

C0

(a) For the distribution to be valid, all probabilities must be nonnegative:

$$r \geq 0, \quad 2r \geq 0, \quad \frac{1}{2} - 2r \geq 0, \quad \frac{1}{4} \geq 0.$$

Hence the valid range is

$$0 \leq r \leq \frac{1}{4}.$$

The sum condition is

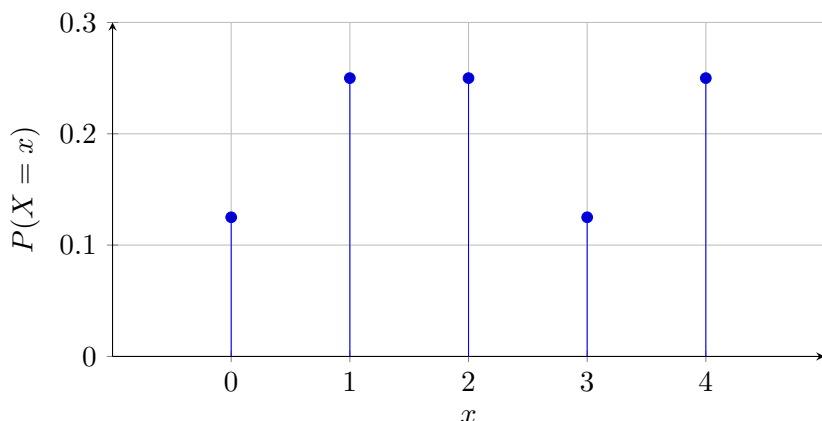
$$r + 2r + \left(\frac{1}{2} - 2r\right) + r + \frac{1}{4} = 1 \Rightarrow 2r + \frac{3}{4} = 1 \Rightarrow r = \frac{1}{8}.$$

Since

$$0 \leq \frac{1}{8} \leq \frac{1}{4},$$

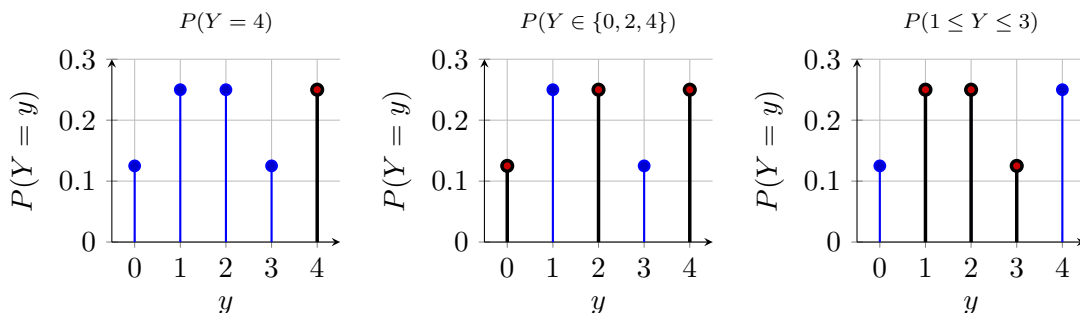
the value $r = \frac{1}{8}$ is valid.

$$P(Y = y) = \{(0, 0.125), (1, 0.25), (2, 0.25), (3, 0.125), (4, 0.25)\}.$$



(b)

$$P(Y = 4) = \frac{1}{4}, \quad P(Y \in \{0, 2, 4\}) = r + \left(\frac{1}{2} - 2r\right) + \frac{1}{4} = \frac{5}{8}, \quad P(1 \leq Y \leq 3) = 2r + \left(\frac{1}{2} - 2r\right) + r = \frac{5}{8}.$$



(c) There is a 25% probability of exactly 4 defects, a 62.5% probability of 0, 2, or 4 defects, and a 62.5% probability of between 1 and 3 defects.

19 Phone Calls to the Office

Problem. Let Z be the number of phone calls to an office in a short period. The probability mass function is

z	0	1	2	3
$P(Z = z)$	$\frac{\lambda}{2}$	λ	$\frac{3\lambda}{2}$	λ

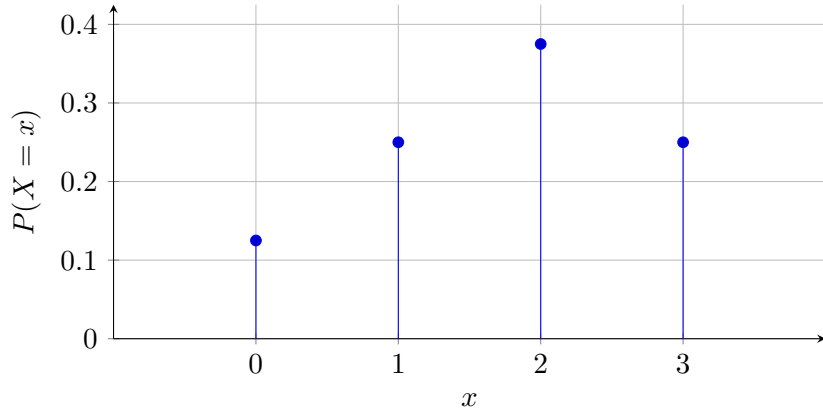
Questions/Tasks. (a) Determine λ . (b) Compute: (i) $P(Z = 2)$, (ii) $P(Z \leq 1)$, and (iii) $P(1 \leq Z \leq 3)$. (c) Interpret these results in terms of percentage probability.

C0

(a) Apply total probability 1:

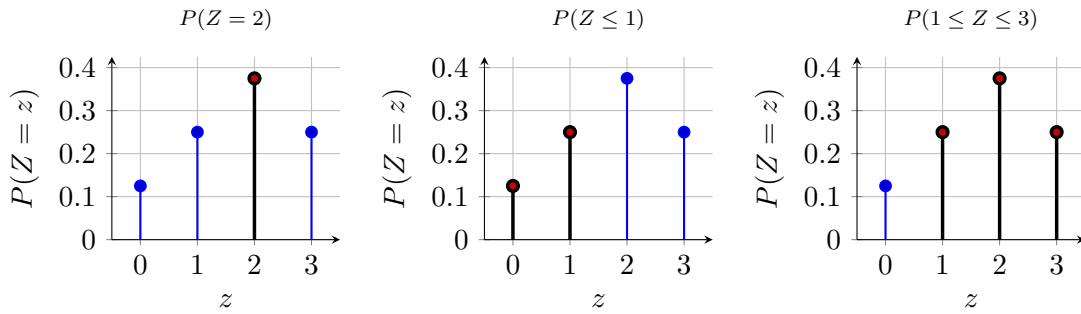
$$\frac{\lambda}{2} + \lambda + \frac{3\lambda}{2} + \lambda = 4\lambda = 1 \Rightarrow \lambda = \frac{1}{4}.$$

$$P(Z = z) = \{(0, 0.125), (1, 0.25), (2, 0.375), (3, 0.25)\}.$$



(b)

$$P(Z = 2) = \frac{3\lambda}{2} = \frac{3}{8}, \quad P(Z \leq 1) = \frac{\lambda}{2} + \lambda = \frac{3}{8}, \quad P(1 \leq Z \leq 3) = \lambda + \frac{3\lambda}{2} + \lambda = \frac{7\lambda}{2} = \frac{7}{8}.$$



(c) There is a 37.5% probability of exactly 2 phone calls, a 37.5% probability of at most 1 phone call, and an 87.5% probability of between 1 and 3 phone calls.

20 Homework Submissions Missing

Problem. A teacher records W , the number of missing homework submissions in a class group.

w	0	1	2	3	4
$P(W = w)$	r	$2r$	$3r$	$\frac{1}{2} - 2r$	$\frac{1}{4}$

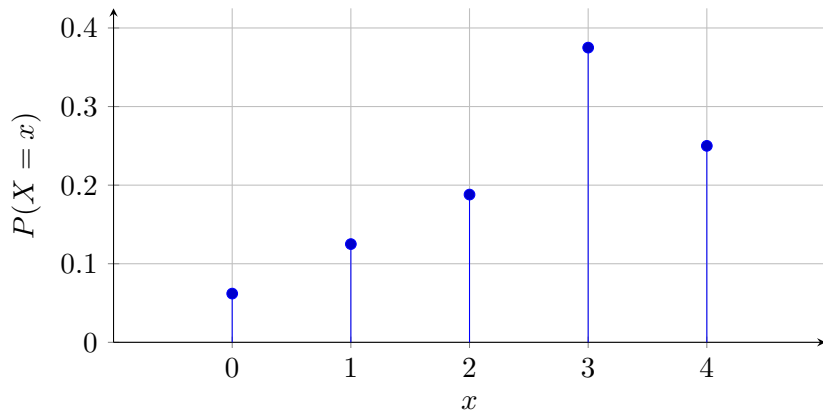
Questions/Tasks. (a) Determine r so this is a valid distribution. (b) Compute: (i) $P(W = 4)$, (ii) $P(W > 2)$, and (iii) $P(1 \leq W \leq 3)$. (c) Interpret these results in terms of percentage probability.

C0

(a) Check total probability:

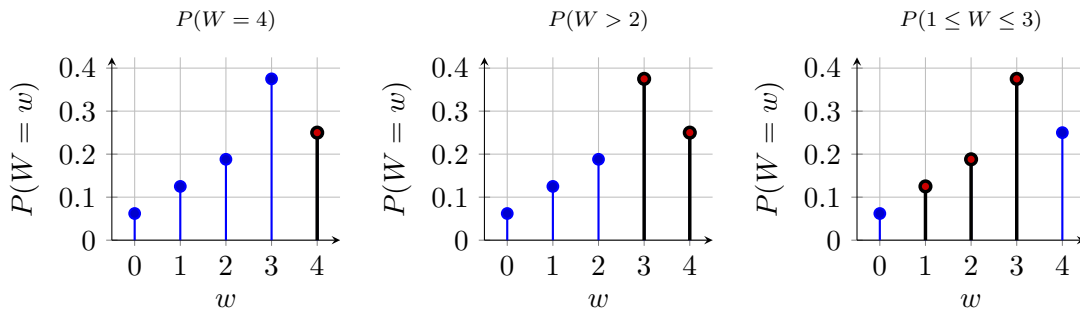
$$r + 2r + 3r + \left(\frac{1}{2} - 2r\right) + \frac{1}{4} = 1 \Rightarrow 4r + \frac{3}{4} = 1 \Rightarrow r = \frac{1}{16}.$$

$$P(W = w) = \{(0, 0.062), (1, 0.125), (2, 0.188), (3, 0.375), (4, 0.25)\}.$$



(b)

$$P(W = 4) = \frac{1}{4}, \quad P(W > 2) = \left(\frac{1}{2} - 2r\right) + \frac{1}{4} = \frac{5}{8}, \quad P(1 \leq W \leq 3) = 2r + 3r + \left(\frac{1}{2} - 2r\right) = \frac{11}{16}.$$



(c) There is a 25% probability of exactly 4 missing submissions, a 62.5% probability of more than 2 missing submissions, and a 68.75% probability of between 1 and 3 missing submissions.

21 Returns Processed at a Counter

Problem. Let T be the number of product returns processed at a counter in a short period. Suppose

t	0	1	2	3	4
$P(T = t)$	g	$2g$	$2g$	g	g

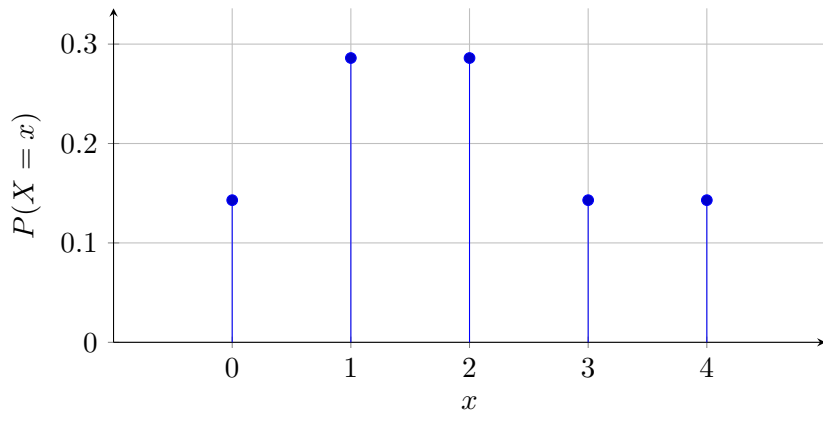
Questions/Tasks. (a) Find g . (b) Compute: (i) $P(T = 2)$, (ii) $P(T \neq 2)$ using a complement, and (iii) $P(1 \leq T \leq 3)$. (c) Interpret these results in terms of percentage probability.

C0

(a) Use the total probability rule:

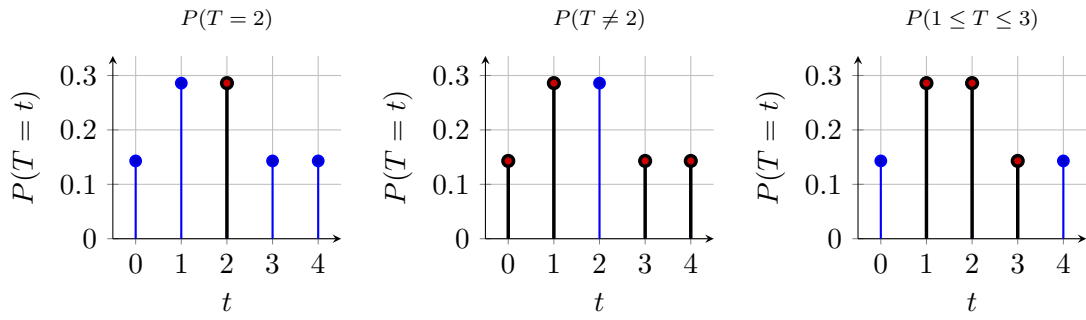
$$g + 2g + 2g + g + g = 7g = 1 \Rightarrow g = \frac{1}{7}.$$

$$P(T = t) = \{(0, 0.143), (1, 0.286), (2, 0.286), (3, 0.143), (4, 0.143)\}.$$



(b)

$$P(T=2) = 2g = \frac{2}{7}, \quad P(T \neq 2) = 1 - 2g = \frac{5}{7}, \quad P(1 \leq T \leq 3) = 2g + 2g + g = 5g = \frac{5}{7}.$$



(c) There is a 28.57% probability of exactly 2 returns, a 71.43% probability of a number of returns different from 2, and a 71.43% probability of between 1 and 3 returns.